

CE2007

Microprocessor-based System Design

Microprocessor Packages, Signals and Interfacing

Part 1

Topics Covered

- Evolution of Packages
 - DIP, TSOP, QFP, PGA, BGA, FCBGA, POP
- Microprocessor Signals
 - Memory interface signals of ARM7TDMI
- Microprocessor Interfacing
 - SRAM pins and timing diagrams
 - Tristate buffers
 - Incompatibility in memory interfacing

Evolution of Packages

Microprocessor Packages

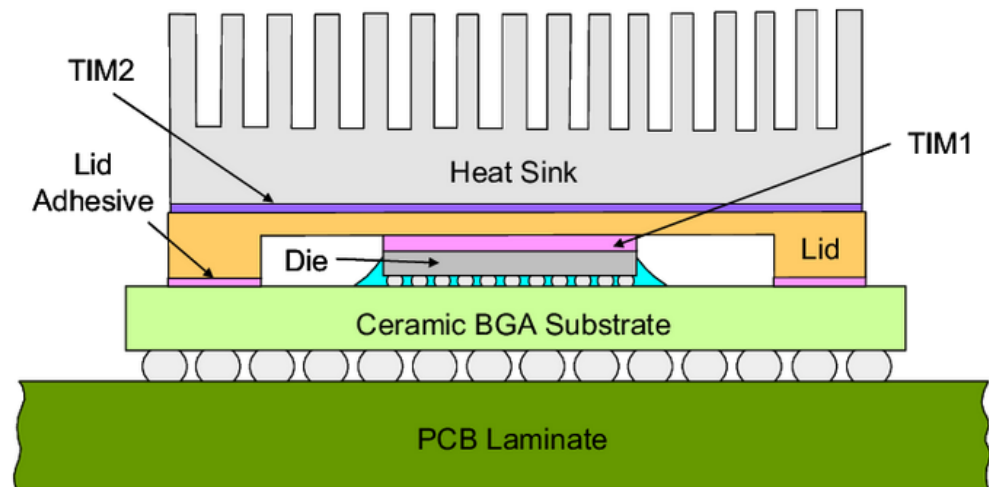
■ Role of Package:

- Prevent damage to the silicon die and the connecting wires
- Enable electrical connection between the die and PCB
- Dissipate heat



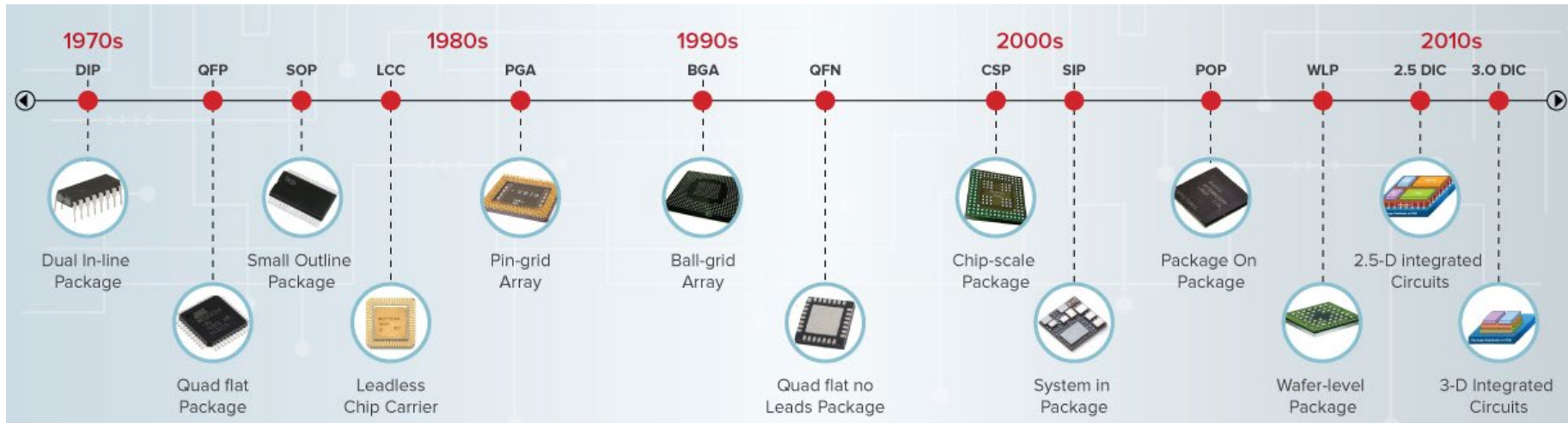
Source:

<http://casualzone.blogspot.sg/2009/12/how-chip-is-made-from-sand-to-silicon.html>



Source: Reliability of a CBGA Microprocessor Package
Incorporating a Decoupling Capacitor Array, 2016

Semiconductor Packaging History and Trends

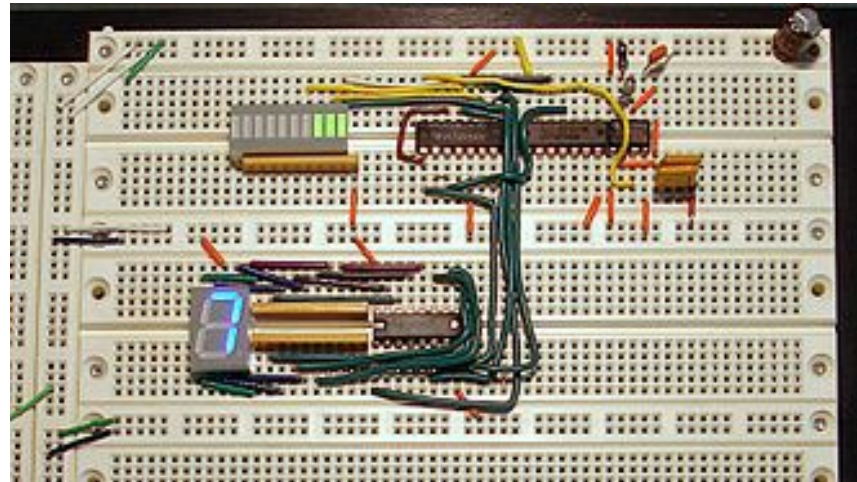
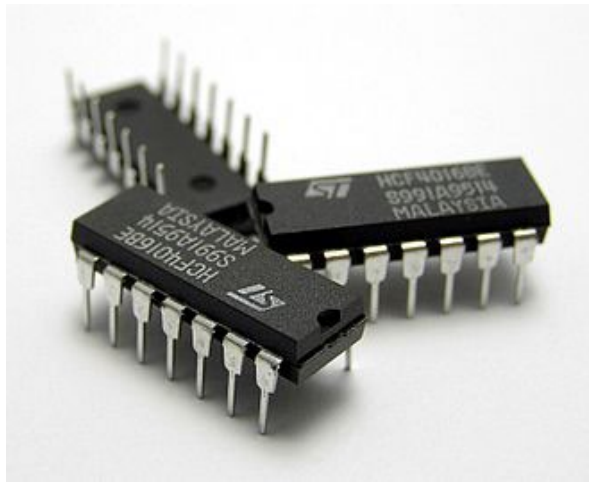


Source: <http://anysilicon.com/semiconductor-packaging-history-trends/>

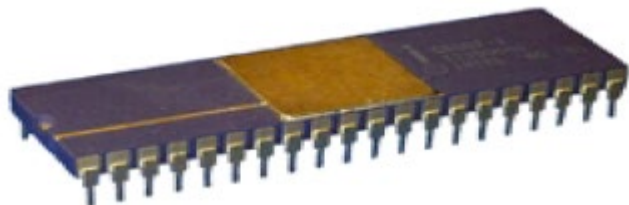
- Addressing high pin counts
 - Modern chips integrate more functionality (computing networking, storage) which require large data or address buses for transferring data in and out of chip
- Accommodating small footprint requirements
 - Driven by miniaturization of mobile phones, laptops, etc.

Evolution of Packages

- Dual In-line Package (DIP) - Early 70's
 - Invented by engineers from Fairchild – 14-lead ceramic DIP
 - Maximum number of pins about 50
 - Examples: Intel 4004, 8080, 8086, 8088



Source: https://en.wikipedia.org/wiki/Dual_in-line_package



40 Lead DIP for 8086 and 8088 microprocessors

Evolution of Packages

- Thin Small Outline Package (TSOP) – 1970-80
 - Surface mount IC package
 - Frequently used for RAM or Flash memory ICs due to their high pin count and small volume
 - Comes in a variety of form factors:
 - Plastic Small-Outline Package (PSOP)
 - Shrink Small-Outline Package (SSOP)
 - Thin-Shrink Small Outline Package (TSSOP)



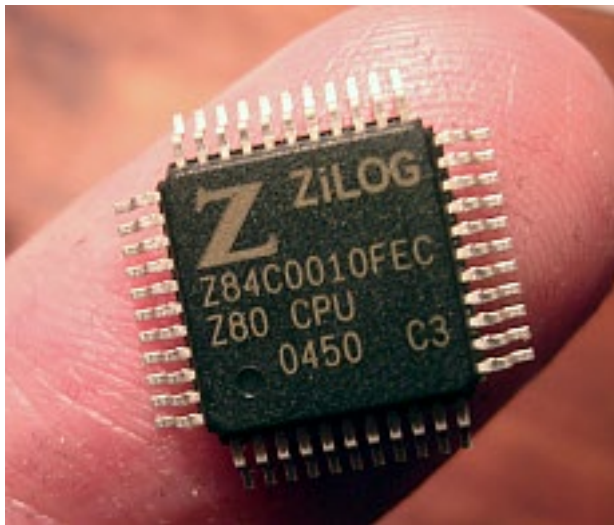
Source:
<http://www.nexperia.com/packages/SOT362-1.html>



Source:
<http://www.wikiwand.com/ru/TSOP>

Evolution of Packages

- Quad Flat Package (QFP) – 1975-80
 - Maximum number of pins about 300
 - Example: Zilog Z80 (44-pin QFP Package)



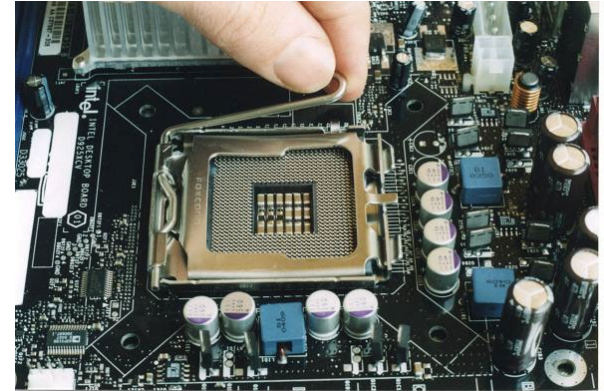
Source:
https://en.wikipedia.org/wiki/Quad_Flat_Package



Source:
<http://shannonstrutz.com/uncategorized/component-packages-totally-explained>

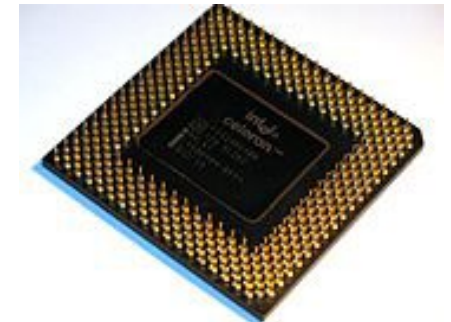
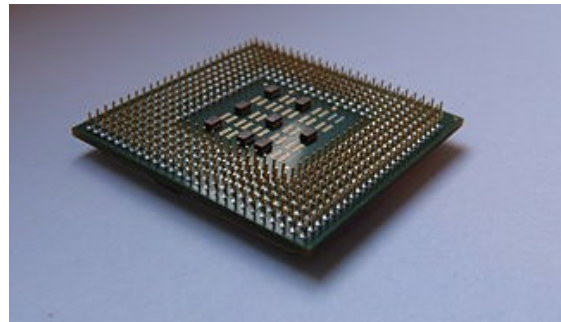
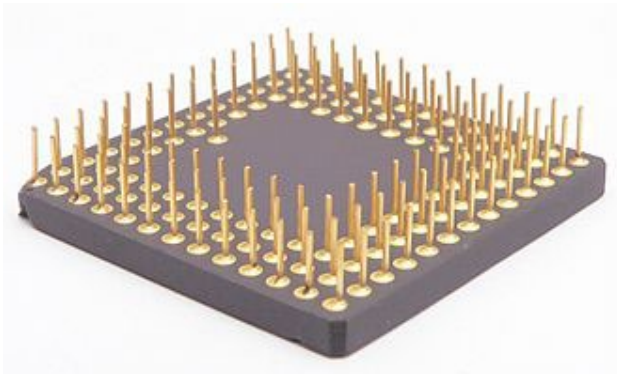
Evolution of Packages

- Pin Grid Array (PGA) – 1985
 - Pins are arranged in a regular array on the underside of the package
 - Can be inserted into a socket
 - Example: Motorola 68020, Pentium 4, Celeron 400



Source:

<http://www.ifitjams.com/ibuild2.htm>



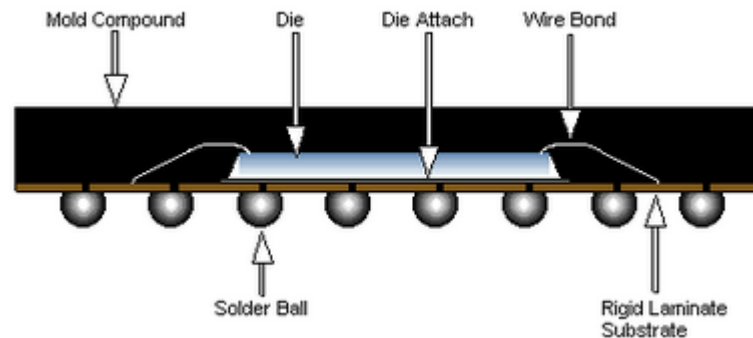
Source: https://en.wikipedia.org/wiki/Pin_grid_array

Evolution of Packages

- Ball Grid Array (BGA) – 1990
 - Overcome structural limits of pins by using a grid array of solder balls
 - Permanently mounted to PCB
 - Chip faces up and is attached to the package via wires



Source:
<http://www.electronicshobby.com/blog/ball-grid-array-bga-package.html>



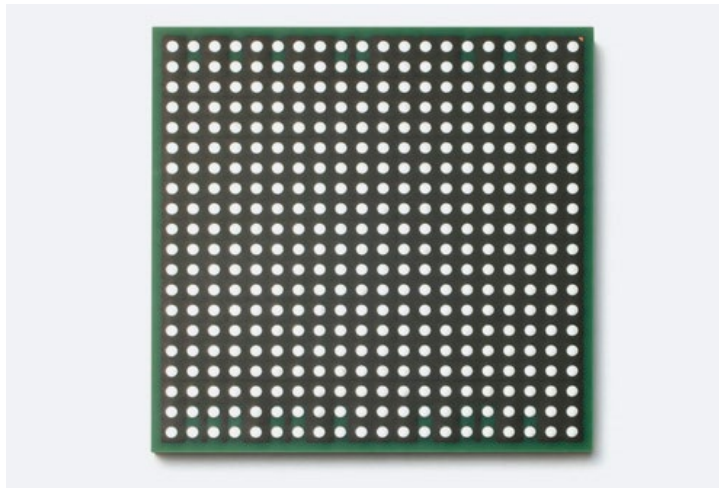
Source:
<http://encyclopedia2.thefreedictionary.com/flip+chip>



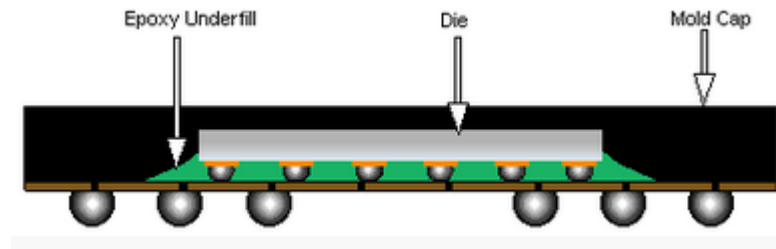
hi-tech-remont.com

Evolution of Packages

- Flip Chip Ball Grid Array (FCBGA) – 2000
 - Chip faces down and is typically attached via solder balls to package
 - Can use whole bottom surface instead of just the perimeter
 - Allows for larger amount of interconnects



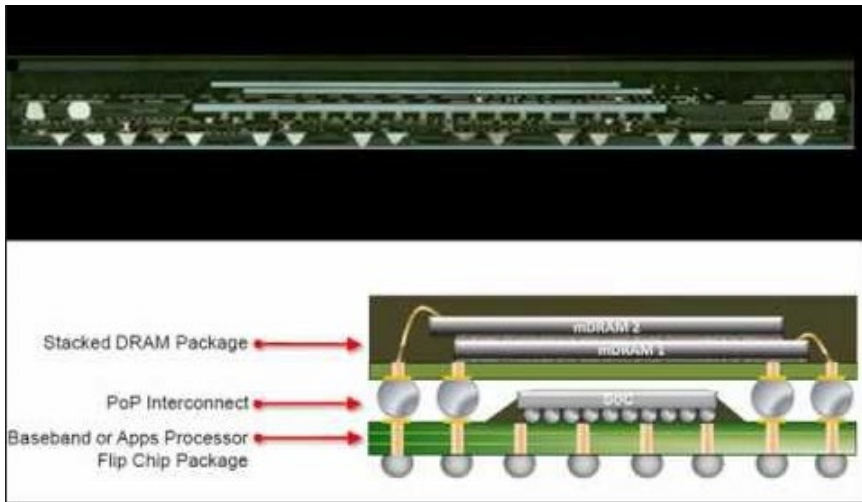
Source:
<http://www.samsungsem.com/global/product/pcb/package-substrate/flip-chip-ball-grid-array/index.jsp>



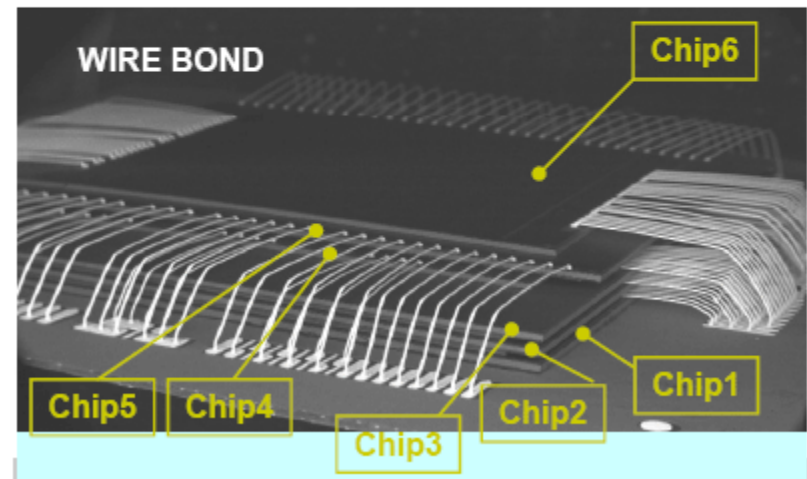
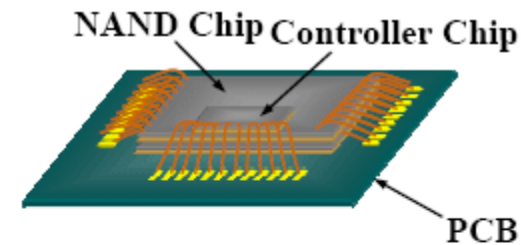
Source:
<http://encyclopedia2.thefreedictionary.com/flip+chip>

Evolution of Packages

- Package on Package (POP) Implementation
 - Reduce form factor
 - Stacking order impacts heat dissipation and interconnect



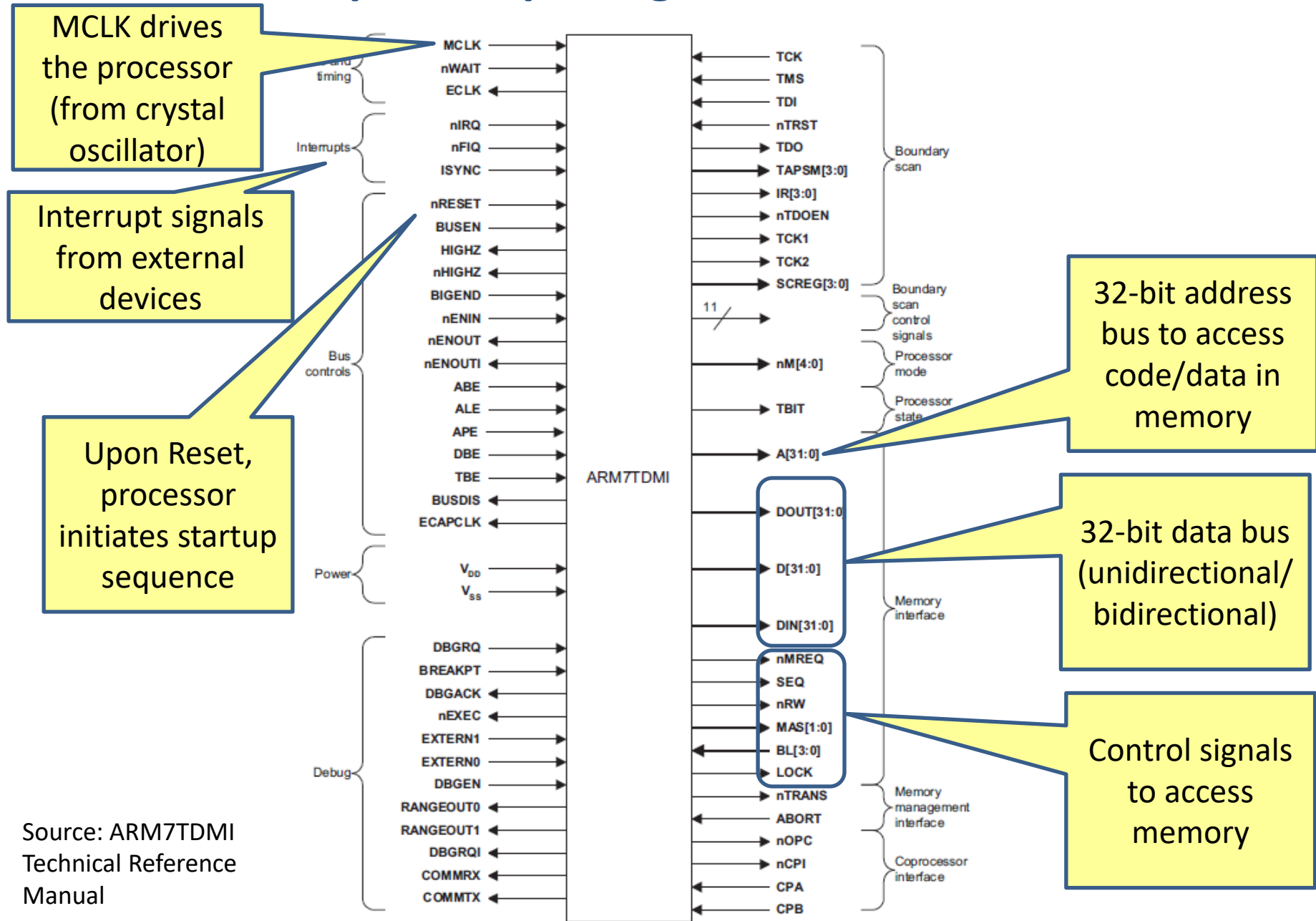
Source:
<https://i.ytimg.com/vi/NljhJqL28eg/hqdefault.jpg>



Source: Tokyo University, 2008

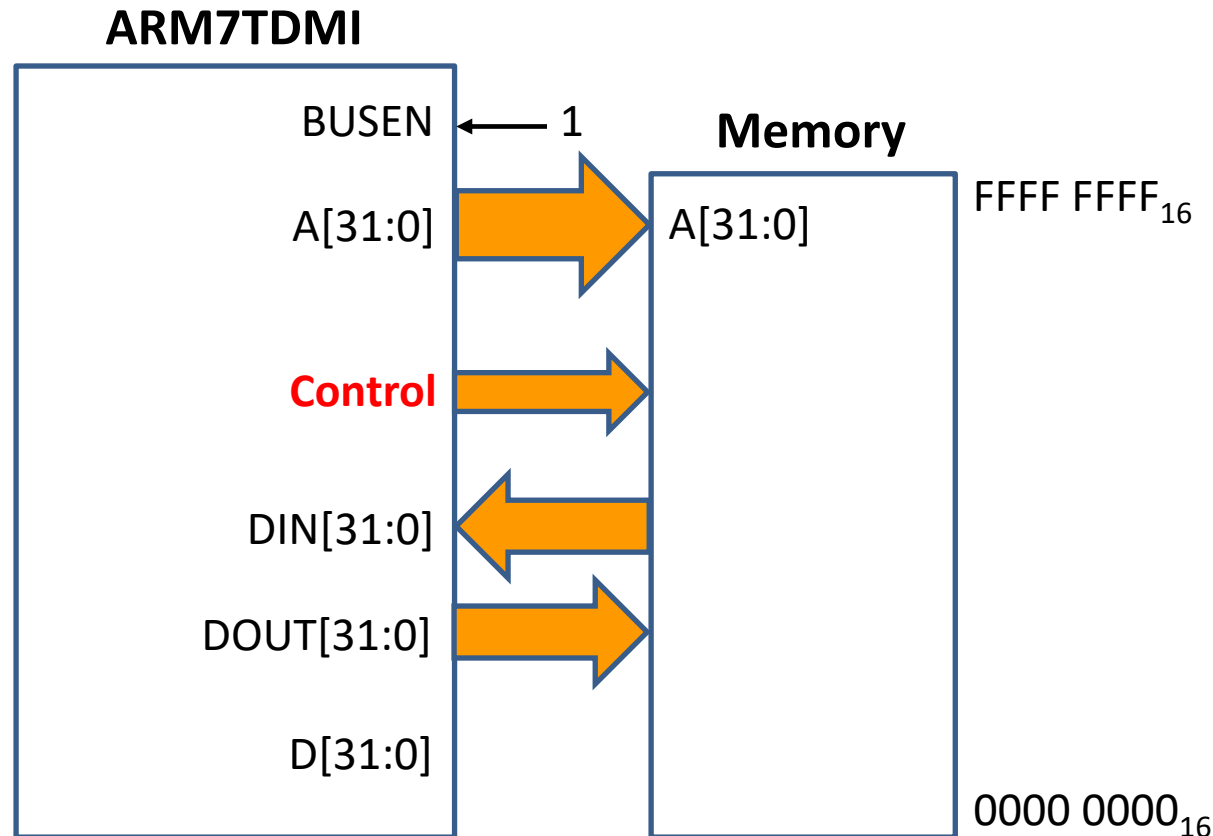
Microprocessor Signals

ARM7TDMI Input/output Signals



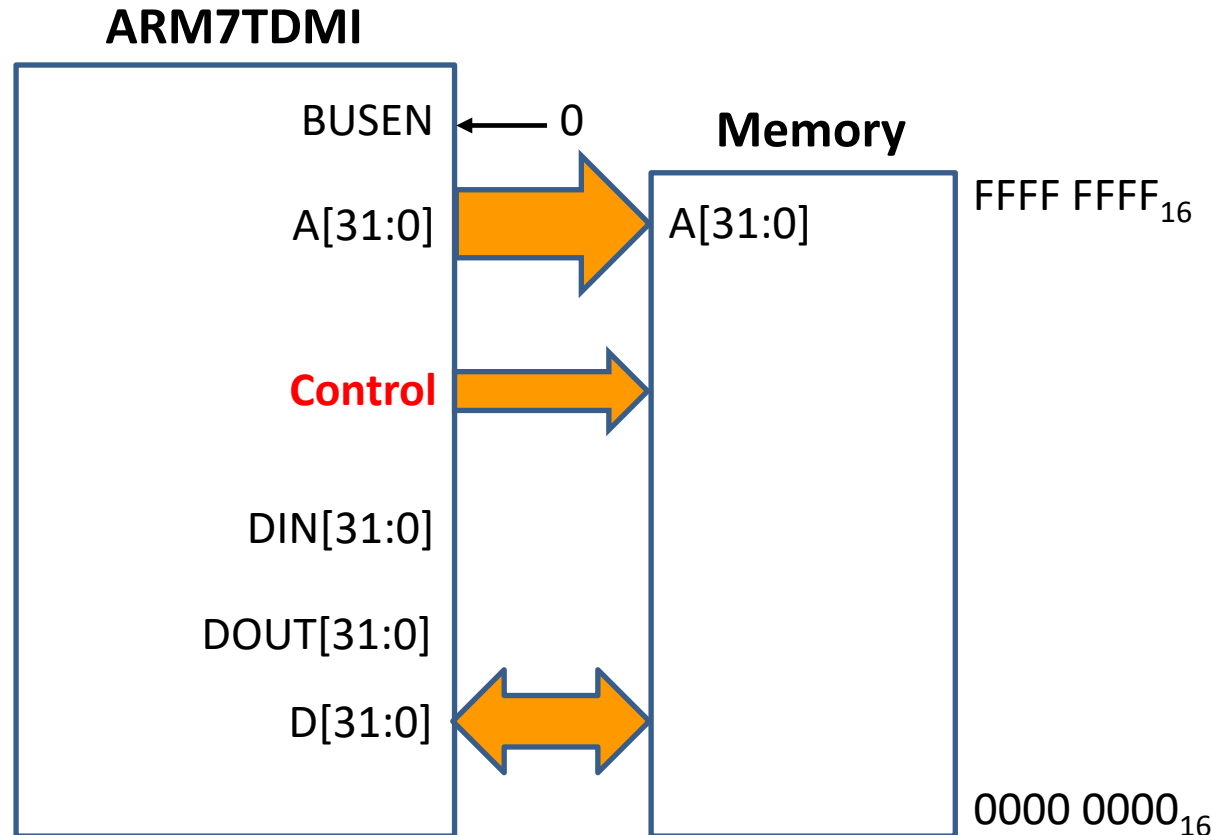
ARM7TDMI Memory Interface Signals

- Clock and Reset
 - MCLK
 - nRESET
- Address Signals
 - A[31:0]
- Data Signals
 - D[31:0]
 - DIN[31:0]
 - DOUT[31:0]
- Control Signals
 - nRW
 - MAS[1:0]



ARM7TDMI Memory Interface Signals

- Clock and Reset
 - MCLK
 - nRESET
- Address Signals
 - A[31:0]
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 - DIN[31:0]
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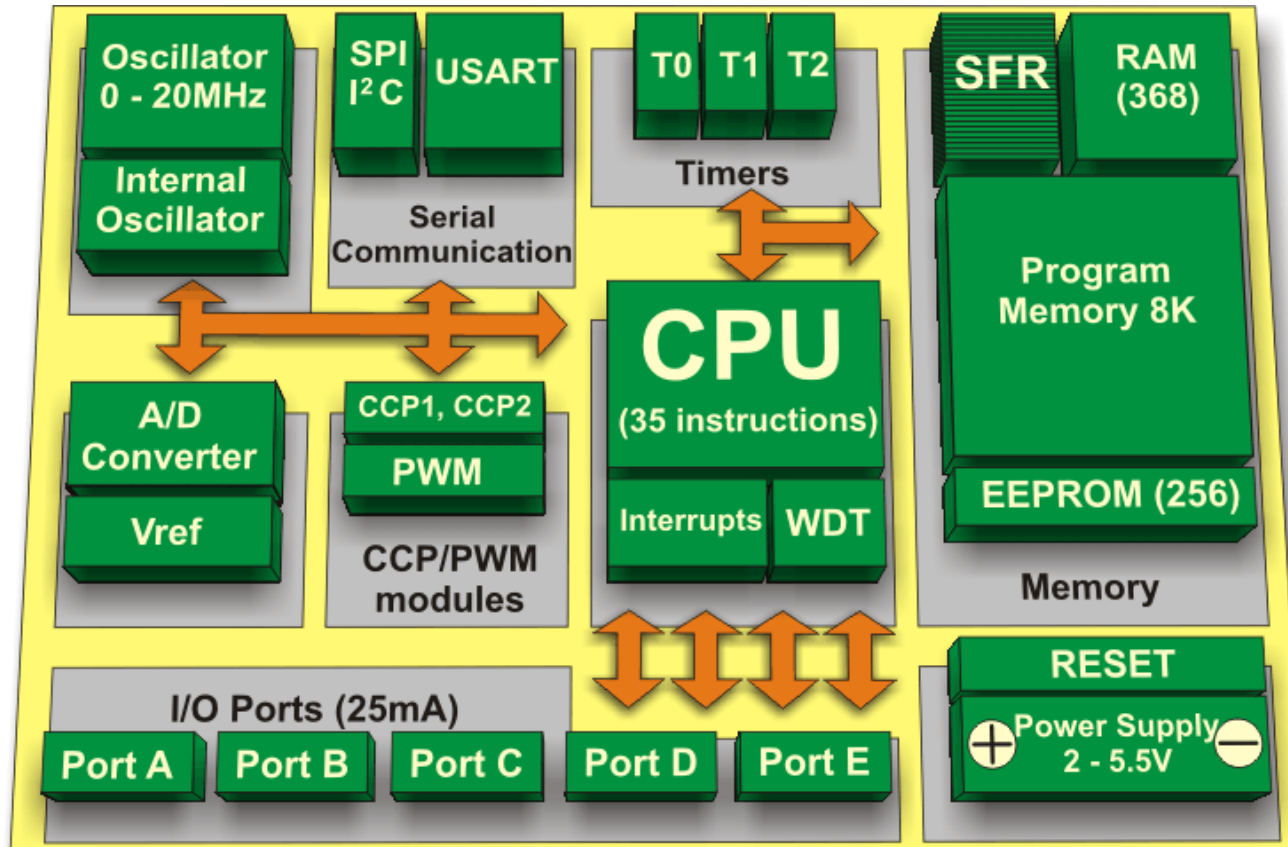


At any one time, only unidirectional or bidirectional data bus is used

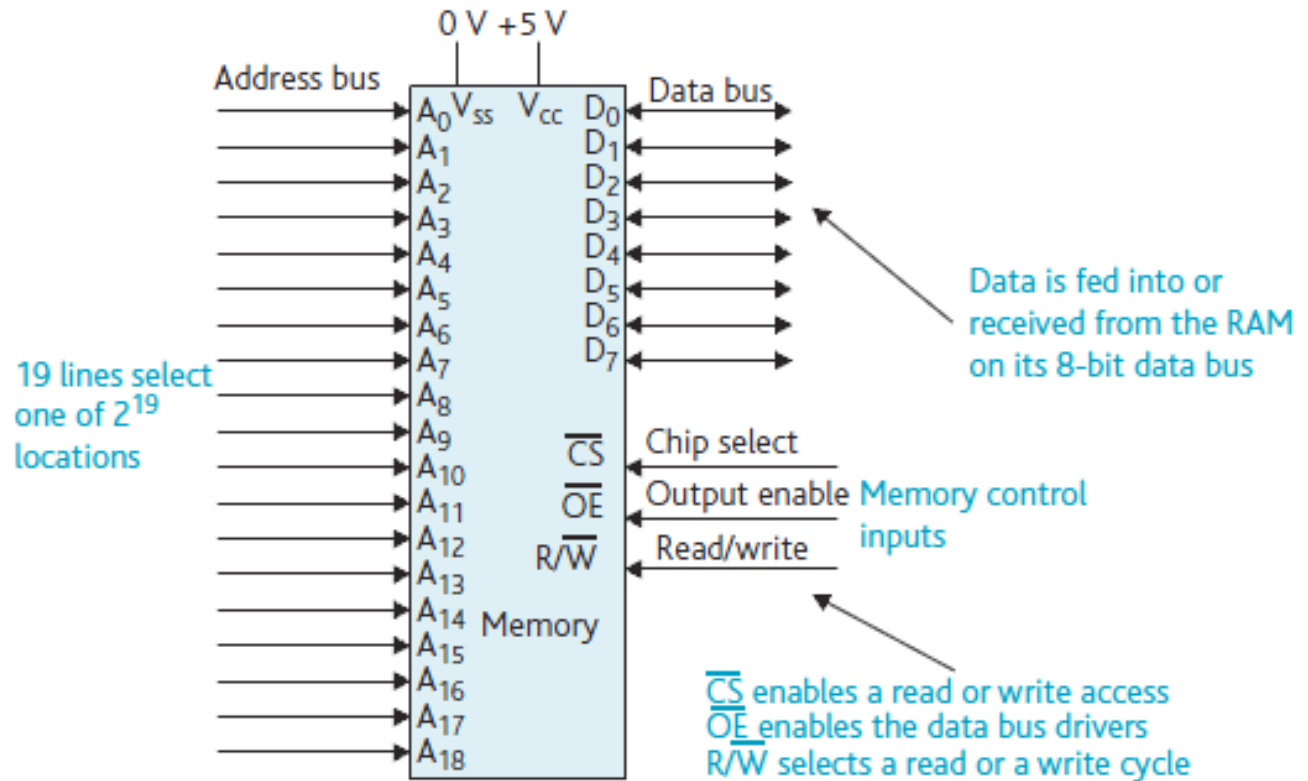
Microprocessor Interfacing

Microcontroller Review

- CPU would already be integrated to all necessary peripherals and delivered as a single-ship solution.



Generic SRAM Pins and Timing Diagrams

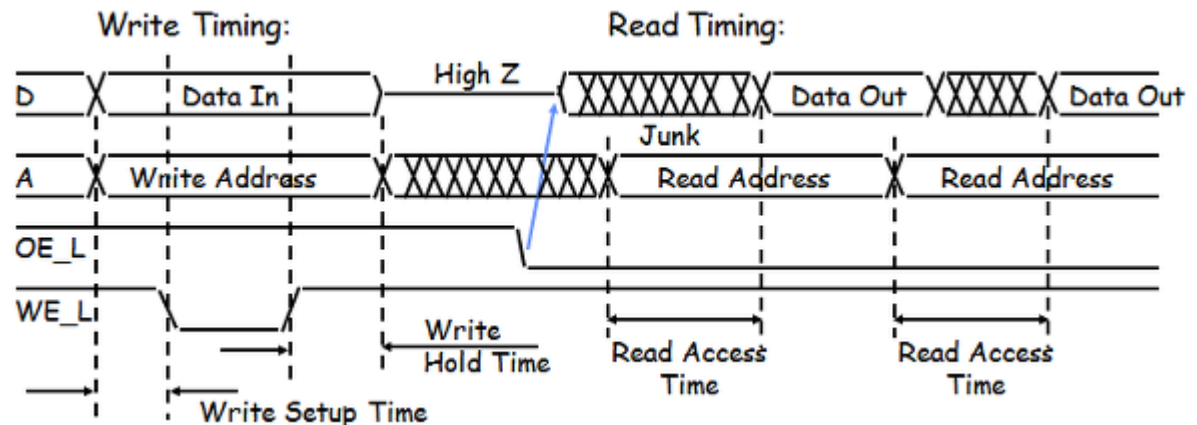


Ports	Description
A[18:0]	Address pins (2 ¹⁹ = 524,288 (512K)) memory locations.
D[7:0]	Data pins (allows a single byte to be transferred a time).
$\overline{CS}$	Chip Select. When low, $\overline{CS}$ selects memory for access.
R/ $\overline{W}$	When high, R/ $\overline{W}$ indicates a read cycle; when low it indicates a write cycle
$\overline{OE}$	Output Enable. When low in read cycle, $\overline{OE}$ allows data to be read from memory and placed on data bus.

Source:
Clements,
"Principle of
Computer
Hardware", 2006

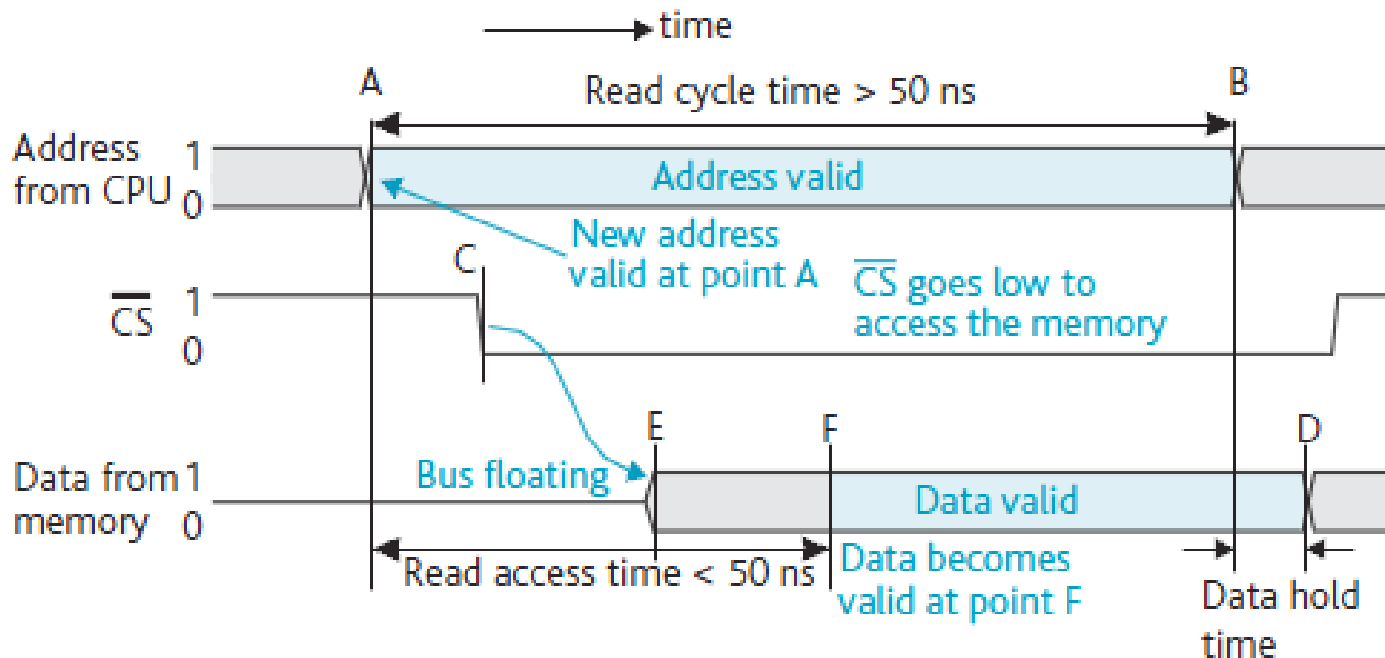
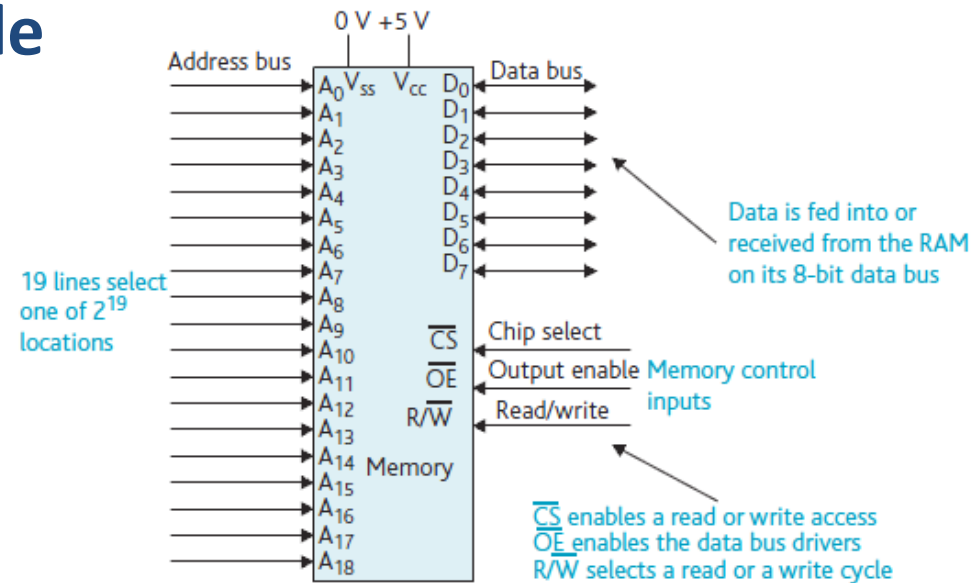
Timing Diagrams

- Cause-and-Effect diagram that illustrates the sequence of actions taking place during a read or write cycle.
- Shows the relationship between information on the address and data buses, and the memory's control inputs.
- Illustrates the state of the signals involved in a memory access as a function of time.
- In a timing diagram,
 - Each signal may be in a 0 or 1 state.
 - Sloping edges indicate change of signal level.
 - Two parallel lines means that some of the lines may be high and some may be low.



Timing Diagram – Read Cycle

What is $\overline{RW}$ and $\overline{OE}$ for read access?



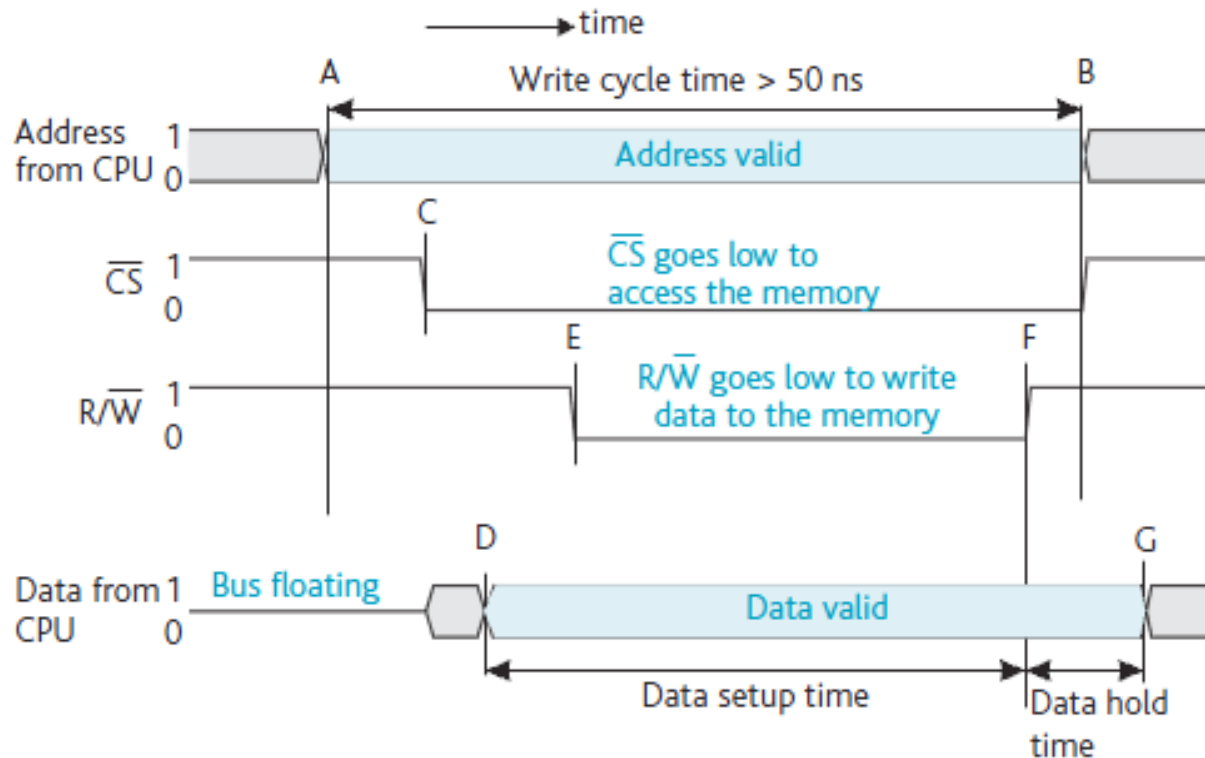
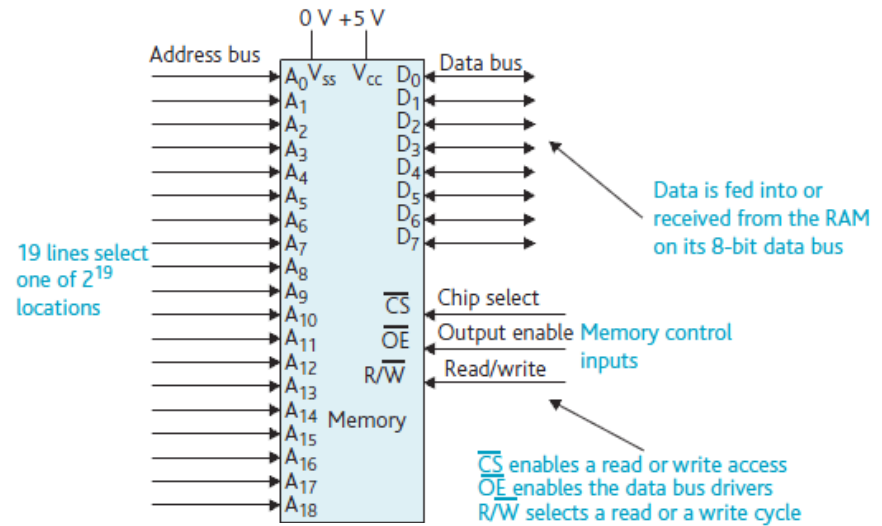
Timing Diagrams – SRAM Read Cycle

- Point A
 - Contents of the address bus have changed from their previous value and are now stable.
 - Between Points A and B, the address bus contains the address of the memory location currently being read from.
 - During this time, the address from the CPU must not change.
 - Time between A and B is known as “Minimum Cycle Time”.
 - Higher-order address lines from the CPU cause a Chip-Select output from the address decoder to be asserted.
- Point C
 - The memory’s active-low chip select input $\overline{CS}$, goes low.
 - Tri-State output is activated.
- Point E
 - Data bus stop floating and data appears at the output terminals.
 - Sufficient time is required for the data contents to be retrieved.

Timing Diagrams – SRAM Read Cycle (Cont.)

- Point F
 - Data is valid.
 - Time between Points A and F is called the chip's read access time.
- Point B
 - Contents of the address bus change.
 - Due to propagation delays, the data at the output pins does not change until a minimum time called "Data Hold Time".

Timing Diagram – Write Cycle



Timing Diagrams – SRAM Write Cycle

- Data from CPU is presented to memory at its data inputs.
- $R/\bar{W}$ is set to low for a WRITE access
- $\bar{CS}$ is asserted
- Data is latched into the memory cell by the rising edge of the $R/\bar{W}$ input.

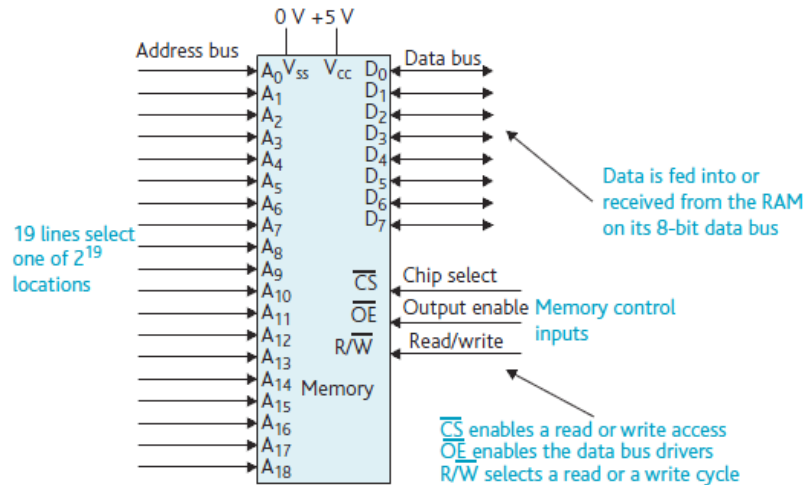
- Critical timing parameters:
 - Write Pulse Width
 - Minimum time for which $R/\bar{W}$ must be low.
 - Data Setup Time
 - Minimum time for which data must be present at the input terminals of a memory before the data is latched in.

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1. $\overline{WE}$ is HIGH for a Read Cycle.
2. The device is continuously selected. $\overline{OE}$, $\overline{CE}$, $\overline{UB}$, or $\overline{LB} = V_{IL}$.
3. Address is valid prior to or coincident with $\overline{OE}$ LOW transition.

Review Question 2

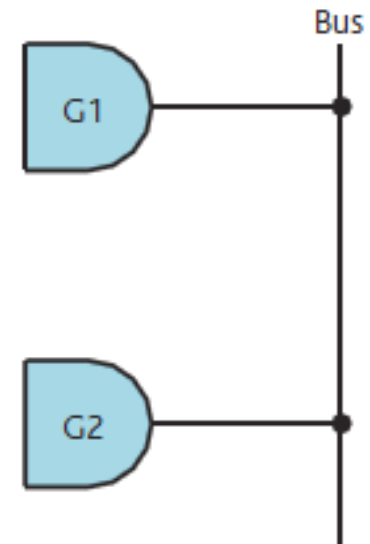
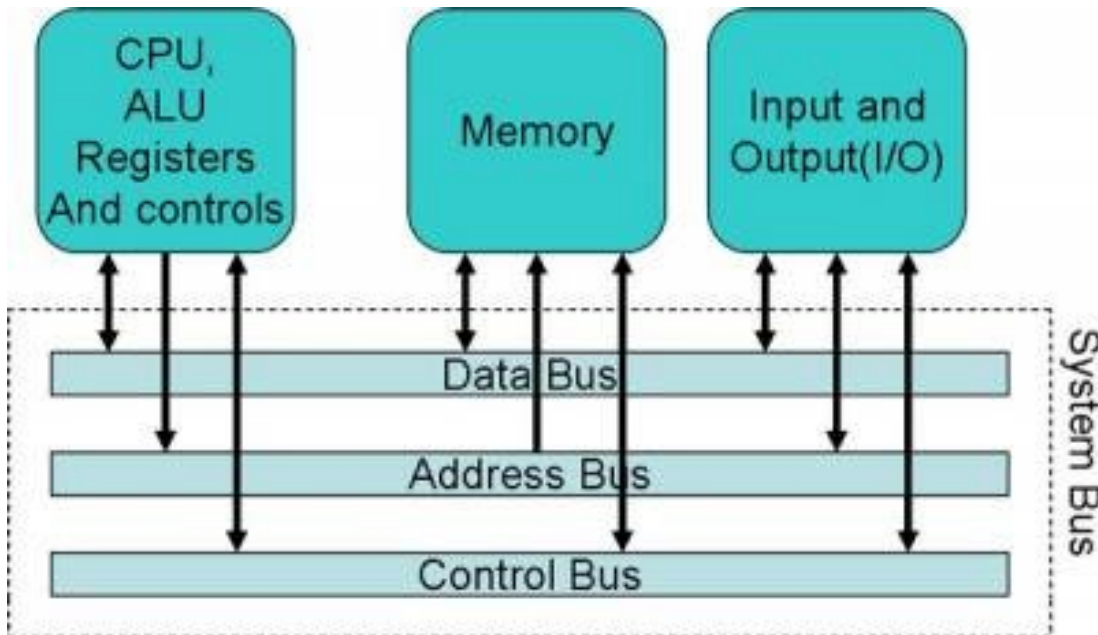
What is $\overline{R\overline{W}}$ and $\overline{O\overline{E}}$ for read access of the SRAM below?



- A. $\overline{R\overline{W}} = 0, \overline{O\overline{E}} = 0$
- B. $\overline{R\overline{W}} = 0, \overline{O\overline{E}} = 1$
- C. $\overline{R\overline{W}} = 1, \overline{O\overline{E}} = 0$
- D. $\overline{R\overline{W}} = 1, \overline{O\overline{E}} = 1$
- E. $\overline{R\overline{W}} = 1, \overline{O\overline{E}} = Hi - Z$

Tristate Buffer

- In a typical system, devices share a common bus. Data can flow onto the bus from one device to another.
- Buses may be unidirectional or bidirectional (data can flow in two directions but not simultaneously).

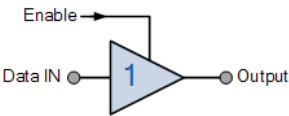


Current flowing through the two output circuits could even destroy the gates.

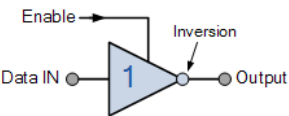
Tristate Buffer

- A logic element with a tristate output has the special property that the output can be in a 0 state, a 1 state, or an unconnected state (high impedance).

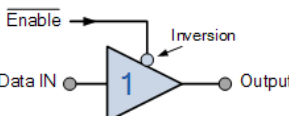
Active "HIGH" Tristate Buffer

Symbol	Truth Table		
 Tri-state Buffer	Enable	IN	OUT
	0	0	Hi-Z
	0	1	Hi-Z
	1	0	0
	1	1	1
Read as Output = Input if Enable is equal to "1"			

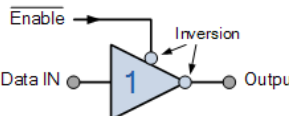
Active "HIGH" Inverting Tristate Buffer

Symbol	Truth Table		
 Inverting Tri-state Buffer	Enable	IN	OUT
	0	0	Hi-Z
	0	1	Hi-Z
	1	0	1
	1	1	0
Read as Output = Inverted Input if Enable equals "1"			

Active "LOW" Tristate Buffer

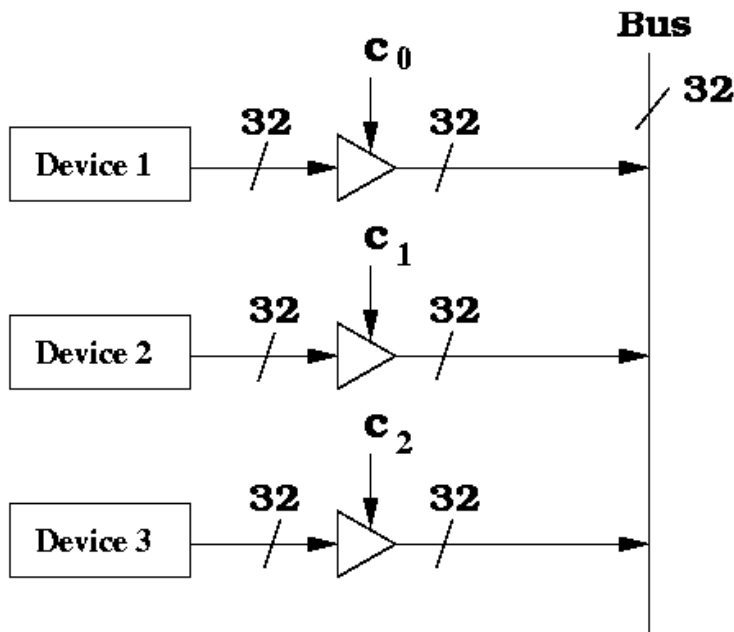
Symbol	Truth Table		
 Tri-state Buffer	Enable	IN	OUT
	0	0	0
	0	1	1
	1	0	Hi-Z
	1	1	Hi-Z
Read as Output = Input if Enable is NOT equal to "1"			

Active "LOW" Inverting Tristate Buffer

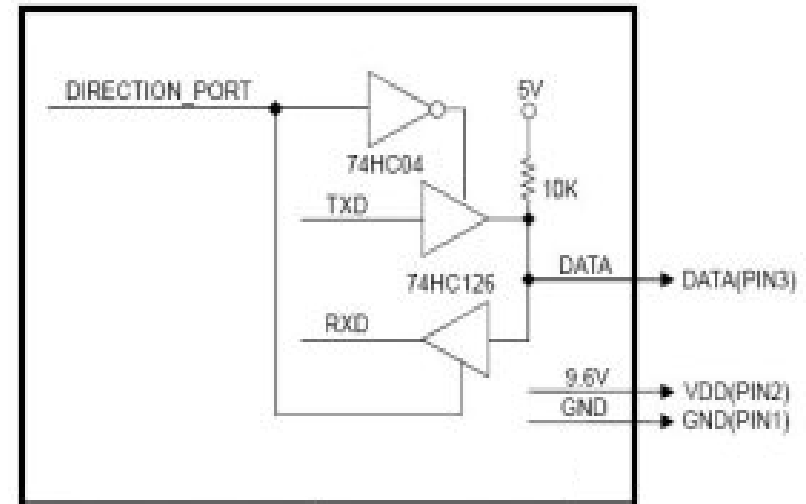
Symbol	Truth Table		
 Inverting Tri-state Buffer	Enable	IN	OUT
	0	0	1
	0	1	0
	1	0	Hi-Z
	1	1	Hi-Z
Read as Output = Inverted Input if Enable is NOT equal to "1"			

Tristate Logic

- Tristate logic disconnects from the bus all those gates not actively engaged in transmitting data.
- In other words, a lot of tristate outputs may be wired to a bus, but only one of them may be actively connected to the bus internally.



Unidirectional data transfer



Bidirectional data transfer

Memory Alignment for Cortex-M1

■ 32-bit Words

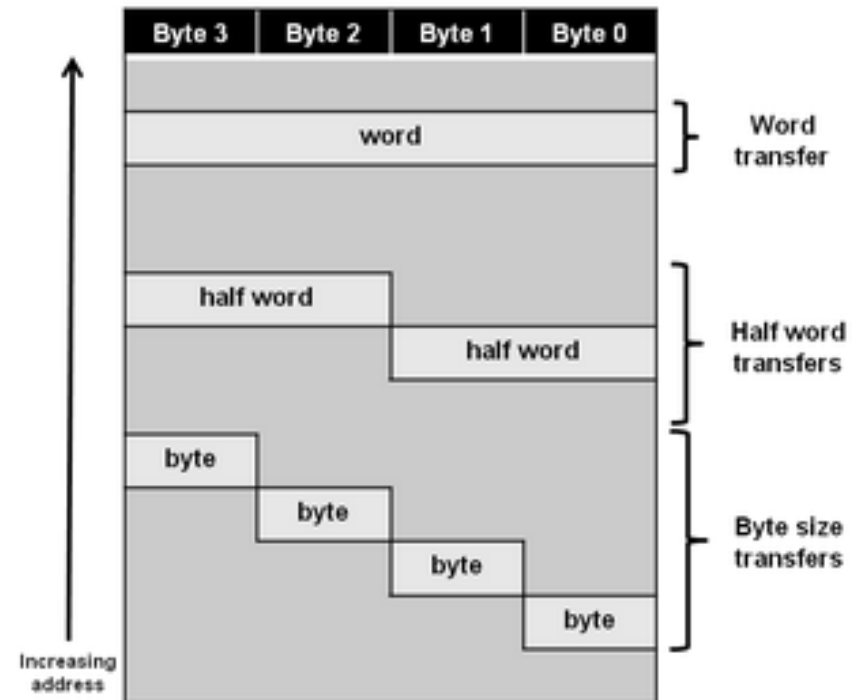
- Address location is aligned to four byte boundaries, hence the lowest two address bits are zero (xxxxxx00b)
- Data are stored as Little Endian (LSB at lower address) by default

■ 16-bit Half-Words

- Address location is aligned to two byte boundaries, hence the lowest address bit is zero (xxxxxxxx0b)

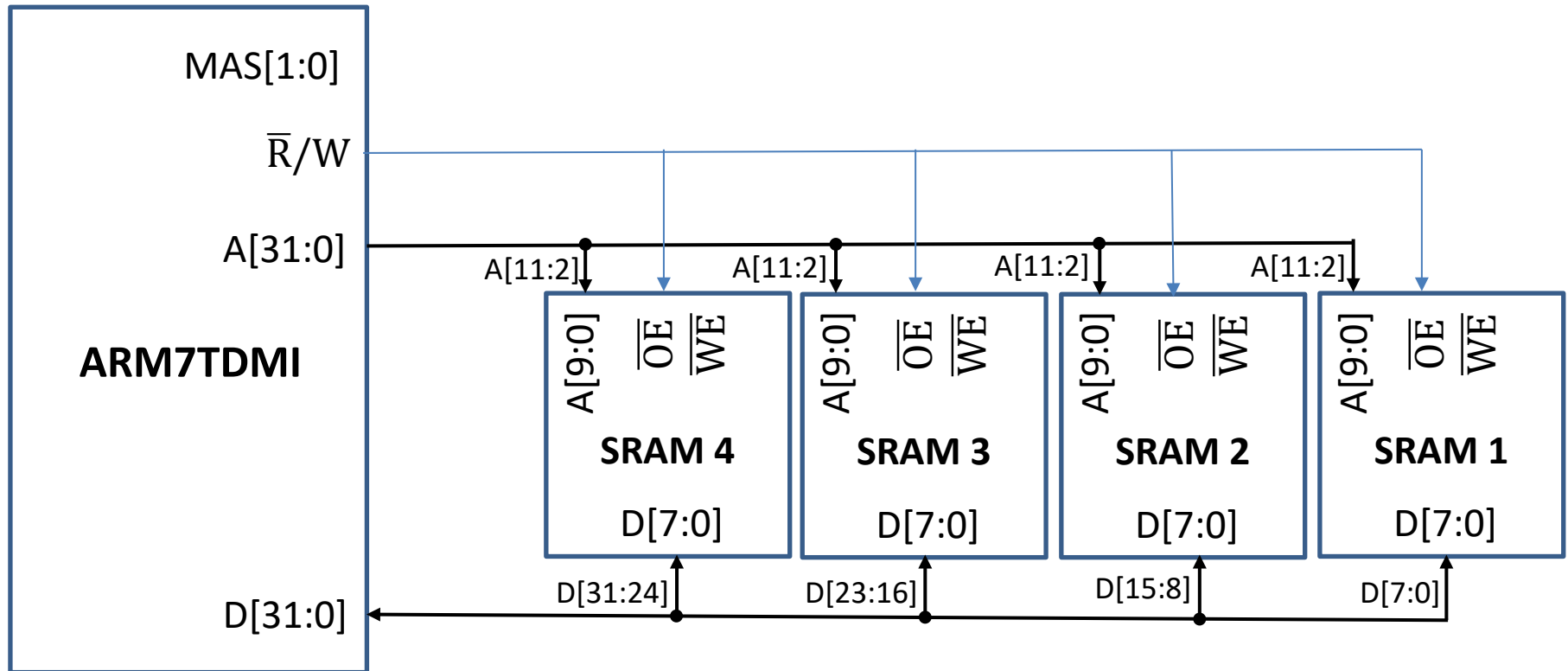
■ 8-bit Bytes

- Can be placed on any byte boundary address

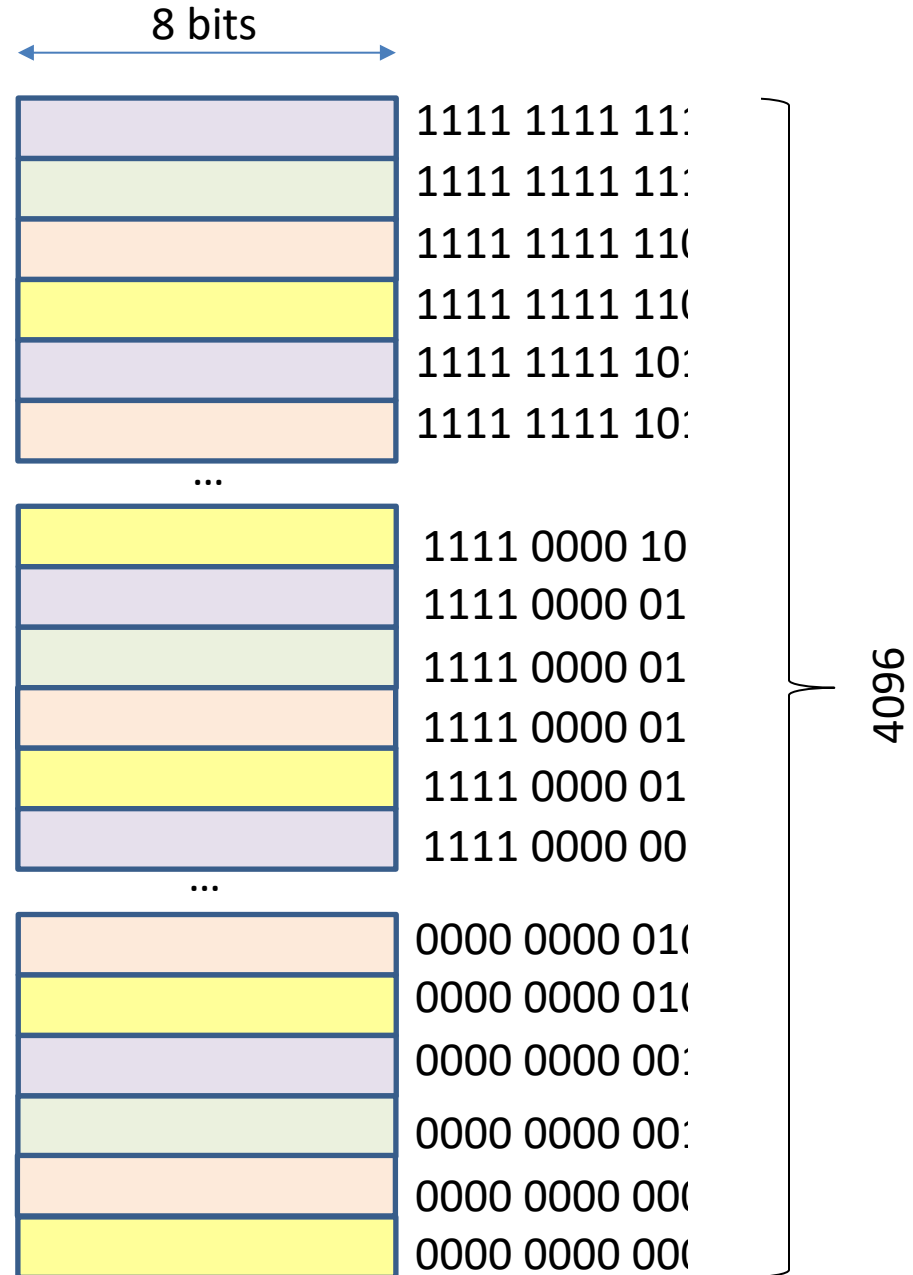
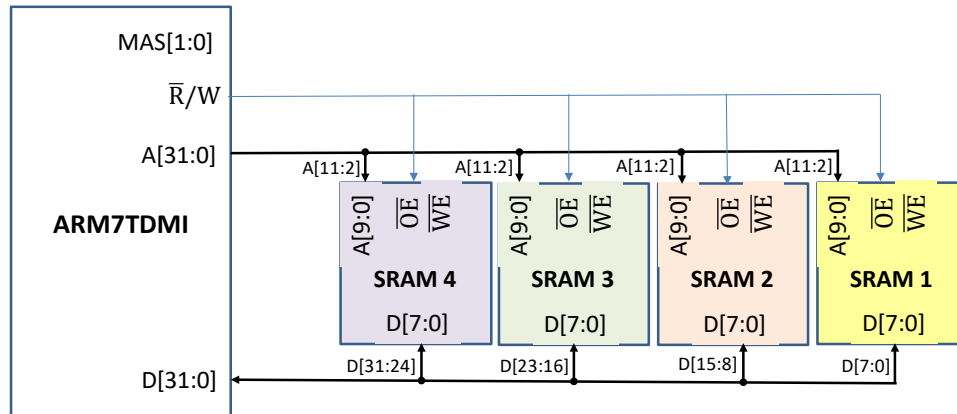


Source: <http://microchipdeveloper.com/32bit:mz-arch-memory-alignment>

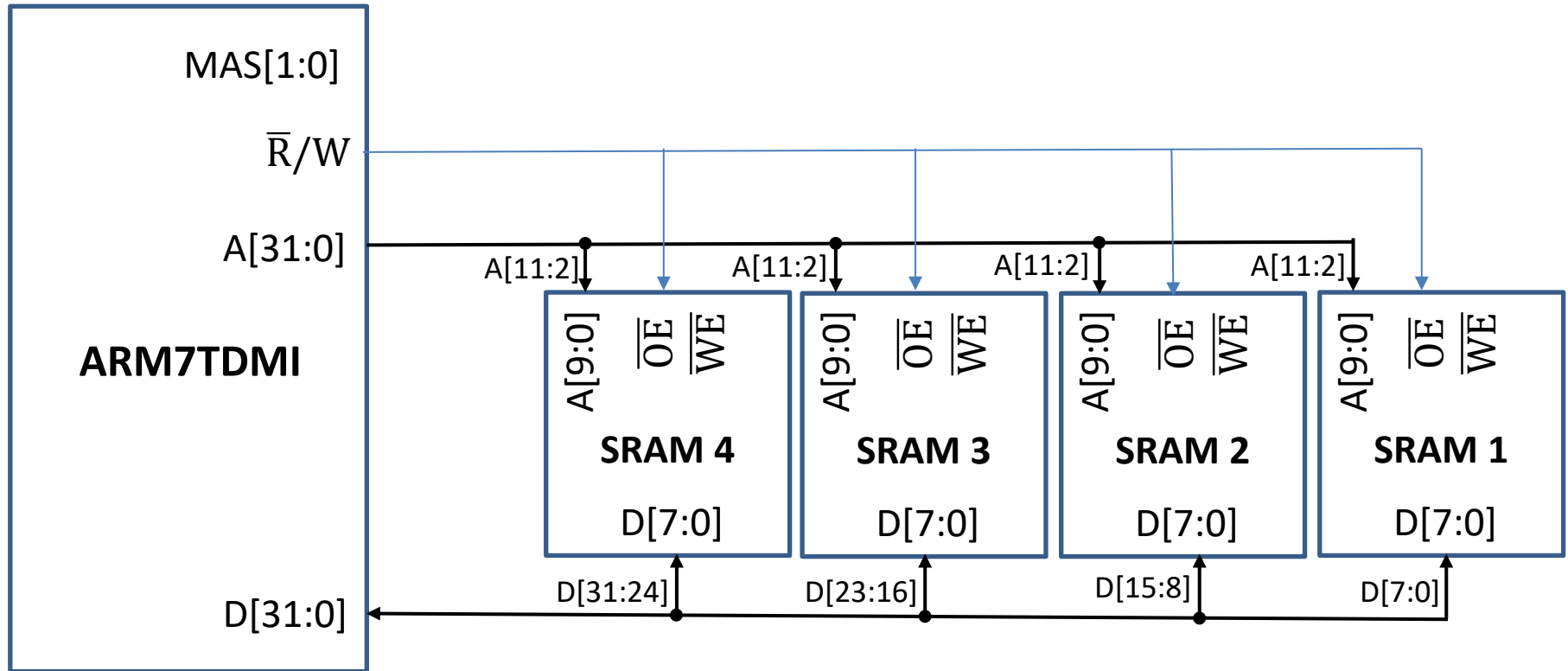
Incompatibility in Memory Interfacing



Address Space



Incompatibility in Memory Interfacing

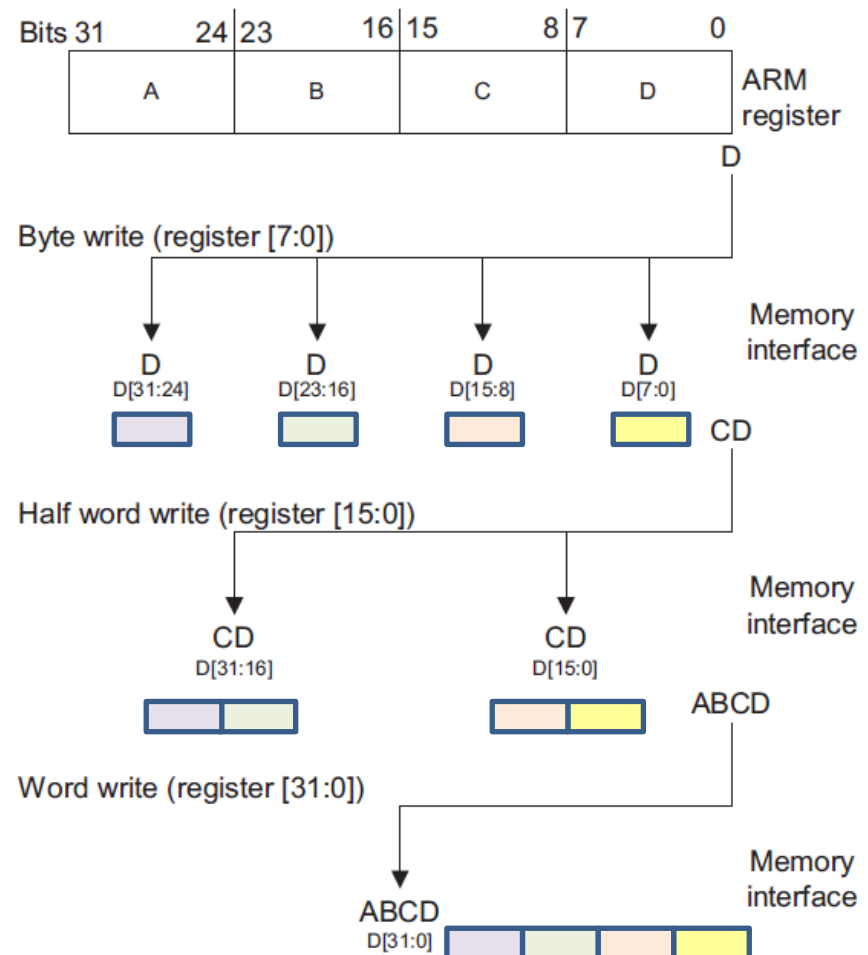
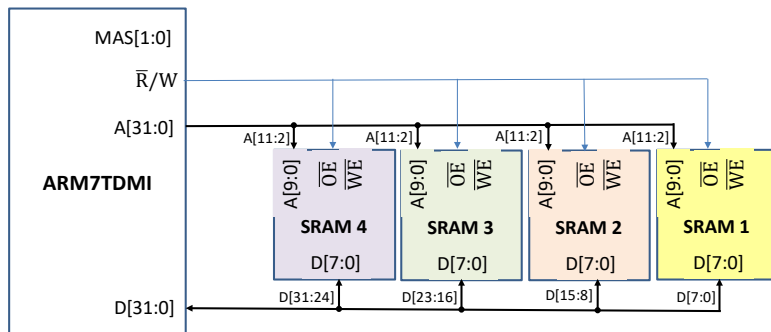


- Read Access
 - Although ARM performs reads of **bytes**, **half-word**, and **words**, memory can ignore the difference and simply supply a word quantity each time
 - When a halfword or byte is read, the 32-bit memory can return the complete 32-bit word, and the processor extracts the valid halfword or byte

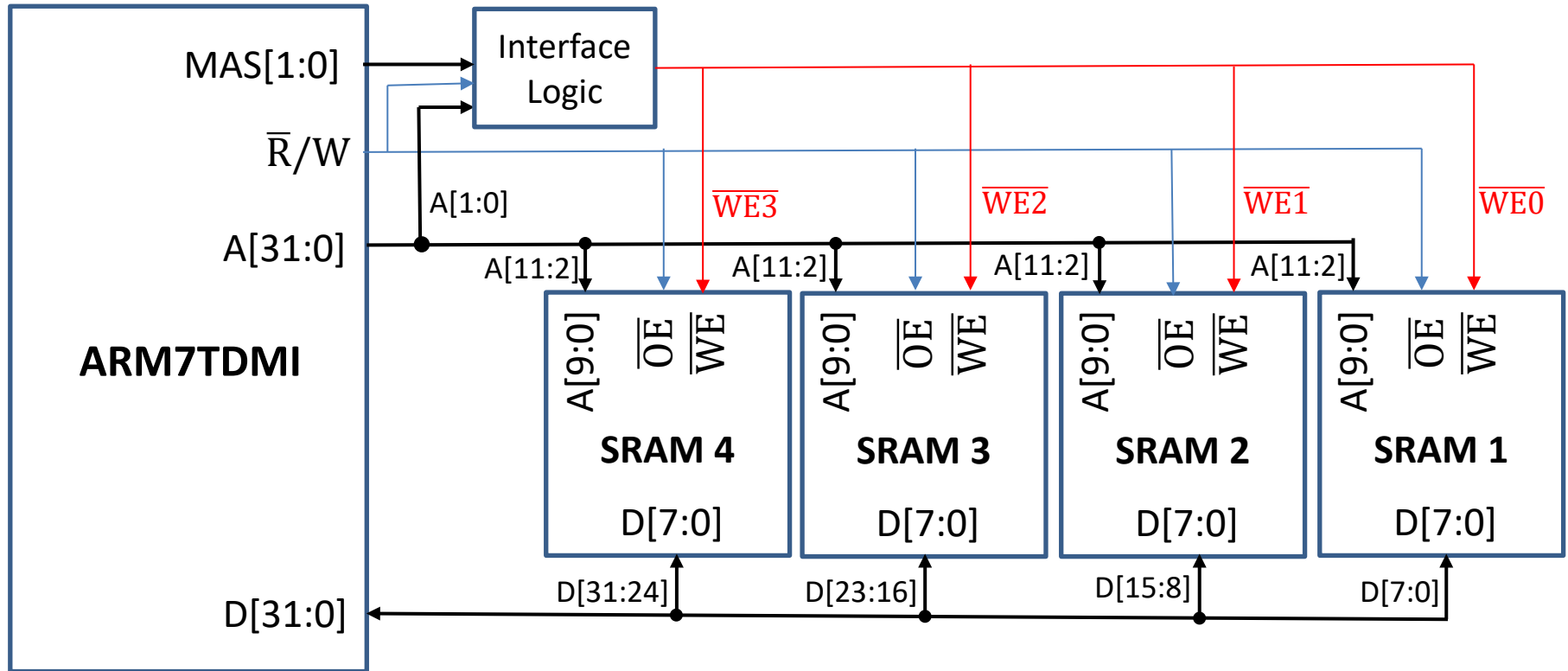
Write Access

- When processor performs a byte or halfword write, data being written is replicated across the data bus
- Memory system use the most convenient way to copy the data

Access type	MAS[1:0]	A[1:0]	Little-endian
Word	10	XX	D[31:0]
Halfword	01	0X	D[15:0]
	01	1X	D[31:16]
Byte	00	00	D[7:0]
	00	01	D[15:8]
	00	10	D[23:16]
	00	11	D[31:24]



Incompatibility in Memory Interfacing



- How do we design the interface logic such that we can write to the correct SRAM chips?
- The bottom two address lines, A[1:0], are used by the interface logic to select the byte or half-word, and they are not connected to memory

Access type	MAS[1:0]	A[1:0]	Little-endian
Word	10	XX	D[31:0]
Halfword	01	0X	D[15:0]
	01	1X	D[31:16]
Byte	00	00	D[7:0]
	00	01	D[15:8]
	00	10	D[23:16]
	00	11	D[31:24]